

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 04 · Constant Acceleration in 1D

With Answers

Questions with full worked solutions

- 40 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 04

Constant Acceleration in 1D

Your classified route

17

QUESTIONS

162

MARKS

5

FAMILIES

11

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 04 | FAMILY 01

Straight-line constant-acceleration SUVAT

WHAT THIS FAMILY ASKS

Choose and apply a scalar constant-acceleration relation in a fixed sign convention.

VARIANT ROUTE

V01 Choose a SUVAT relation

V02 Distance in the n th second or sub-interval

V03 Recover a parameter from average speed

4 classified questions

37 total marks

January 2026 Q5 — complete original question

5. The points O , A , B and C are on a rough horizontal surface where $OABC$ is a straight line.

A particle is projected horizontally from O and slides across the surface, passing through A , B and C in that order.

The particle passes through

- A with speed $U \text{ m s}^{-1}$
- B with speed $\frac{U}{2} \text{ m s}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

It takes 0.75 s for the particle to move from A to B .

- (a) Show that $U = 2.1$ to 2 significant figures.

(6)

The particle passes through C with speed $\frac{U}{3} \text{ m s}^{-1}$

- (b) Find the time taken to move from B to C .

(3)

- (c) Find the distance AC .

(3)

January 2026 Q5 — detailed solution

Choose the right SUVAT relation

ALL
12 marks**Find the acceleration from friction**

On a horizontal surface, $R = mg$ and the friction is $F = \mu R = \frac{1}{7}mg$. The friction is the only horizontal force, so the acceleration is opposite the motion with magnitude $\frac{F}{m} = \frac{g}{7} = 1.4 \text{ m s}^{-2}$.

Use A to B to find U

From A to B, $u = U$, $v = \frac{U}{2}$, $a = -\frac{g}{7}$ and $t = 0.75$. Using $v = u + at$: $\frac{U}{2} = U - \frac{g}{7}(0.75)$.

Show the value of U

With $g = 9.8$, $\frac{g}{7}(0.75) = 1.05$, so $\frac{U}{2} = U - 1.05$. Hence $U = 2.10 \text{ m s}^{-1}$, which is 2.1 m s^{-1} to 2 significant figures.

Find the time from B to C

From B to C, the speed falls from $\frac{U}{2}$ to $\frac{U}{3}$ with the same acceleration -1.4 . Thus $\frac{U}{3} = \frac{U}{2} - 1.4t$. With $U = 2.1$, this gives $0.7 = 1.05 - 1.4t$, so $t = 0.25 \text{ s}$.

Find the distance AC

Use average speed on each section. $AB = \frac{U + U/2}{2}(0.75) = 1.18125 \text{ m}$ and $BC = \frac{U/2 + U/3}{2}(0.25) = 0.21875 \text{ m}$. Therefore $AC = 1.18125 + 0.21875 = 1.4 \text{ m}$.

Final answer (a) $U = 2.1 \text{ m s}^{-1}$ to 2 s.f.. (b) 0.25 s . (c) $AC = 1.4 \text{ m}$.

Summer 2021 Q2 — complete original question

2. A car moves along a straight horizontal road with constant acceleration $a \text{ m s}^{-2}$ where $a > 0$

The car is modelled as a particle.

At time $t = 0$, the car passes point A and is moving with speed $u \text{ m s}^{-1}$

In the first three seconds after passing A the car travels 20 m.

In the fourth second after passing A the car travels 10 m.

The speed of the car as it passes point B is 20 m s^{-1}

Find the time taken for the car to travel from A to B .

(8)

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Summer 2021 Q2 — detailed solutionDistance in the n th second**ALL**
8 marks**Use the first three seconds**

Using $s = ut + \frac{1}{2}at^2$, the first 3 seconds give $20 = 3u + \frac{1}{2}a(3^2) = 3u + \frac{9}{2}a$.

Use the fourth second

The distance in the fourth second is $u + a(4 - \frac{1}{2}) = u + \frac{7}{2}a$. Hence $10 = u + \frac{7}{2}a$.

Solve for a and u

From $10 = u + \frac{7}{2}a$, $u = 10 - \frac{7}{2}a$. Substitute into $20 = 3u + \frac{9}{2}a$: $20 = 30 - \frac{21}{2}a + \frac{9}{2}a = 30 - 6a$. Thus $a = \frac{5}{3}$.

Find u

$$u = 10 - \frac{7}{2} \cdot \frac{5}{3} = 10 - \frac{35}{6} = \frac{25}{6}.$$

Use the speed at B

At B, speed is 20. Use $v = u + at$: $20 = \frac{25}{6} + \frac{5}{3}t$.

Find the time

$$\frac{5}{3}t = 20 - \frac{25}{6} = \frac{95}{6}, \text{ so } t = \frac{95}{10} = 9.5 \text{ s.}$$

Final answer $t = 9.5 \text{ s.}$

Summer 2022 Q2 — complete original question

2. A motorbike is moving with constant acceleration along a straight horizontal road.

The motorbike passes a point P and 10 seconds later passes a point Q .

The speed of the motorbike as it passes Q is 28 m s^{-1}

Given that $PQ = 220 \text{ m}$,

- (a) find the acceleration of the motorbike,

(3)

- (b) find the distance travelled by the motorbike during the fifth second after passing P

(4)

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Summer 2022 Q2 — detailed solutionDistance in the n th second**ALL**
7 marks**Find the speed at P**

Let the speed at P be u . Over the 10 seconds from P to Q, the speed changes uniformly from u to 28, and the distance is 220 m.

Use average speed

$$220 = \frac{u + 28}{2} \cdot 10 = 5(u + 28). \text{ Hence } u + 28 = 44 \text{ and } u = 16 \text{ m s}^{-1}.$$

Find the acceleration

Use $v = u + at$: $28 = 16 + 10a$, so $a = 1.2 \text{ m s}^{-2}$.

Write the distance from P after t seconds

$$s = 16t + \frac{1}{2}(1.2)t^2 = 16t + 0.6t^2.$$

Find the distance in the fifth second

The fifth second is from $t = 4$ to $t = 5$. $s(5) = 16(5) + 0.6(25) = 95$ and $s(4) = 16(4) + 0.6(16) = 73.6$.

Subtract

Distance in the fifth second = $95 - 73.6 = 21.4 \text{ m}$.

Final answer (a) $a = 1.2 \text{ m s}^{-2}$. (b) 21.4 m.

January 2024 Q3 — complete original question

3. A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed $u \text{ m s}^{-1}$ and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

- (a) Show that $u = 12$ (2)

- (b) Find the time taken for the van to travel from A to the midpoint of AB . (5)

The van has mass 1200 kg.

During its motion the van experiences a constant resistive force of magnitude 260 N

- (c) Find the magnitude of the driving force exerted by the engine of the van as it travels from A to B . (3)

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January 2024 Q3 — detailed solution

Average-speed parameter

ALL
10 marks**Use average speed from A to B**

With constant acceleration, distance equals average speed times time. So $400 = \frac{u + 28}{2} \cdot 20$.

Show u is 12

$400 = 10(u + 28)$, so $u + 28 = 40$ and $u = 12$.

Find the acceleration

Use $v = u + at$: $28 = 12 + 20a$, hence $a = 0.8 \text{ m s}^{-2}$.

Use the midpoint distance

The midpoint of AB is 200 m from A. Use $s = ut + \frac{1}{2}at^2$: $200 = 12t + \frac{1}{2}(0.8)t^2$.

Solve for the time to the midpoint

$200 = 12t + 0.4t^2$. The positive solution is $t = 5\sqrt{29} - 15 = 11.9258 \dots \text{ s}$.

Use Newtons second law for the van

Let the driving force be D . The resultant forward force is $D - 260$. Since $ma = 1200(0.8) = 960$, $D - 260 = 960$.

Find the driving force

$D = 1220 \text{ N}$.

Final answer (a) $u = 12$. (b) $t = 5\sqrt{29} - 15 \approx 11.9 \text{ s}$. (c) driving force = 1220 N.

TOPIC 04 | FAMILY 02

Vertical motion under gravity

WHAT THIS FAMILY ASKS

Apply a single upward/downward sign convention with acceleration of magnitude g .

VARIANT ROUTE

V01 Time or speed at ground impact

V02 Duration above a level

5 classified questions

43 total marks

January 2021 Q1 — complete original question

1. A small stone is projected vertically upwards with speed 20 ms^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

- (a) the speed of the stone at the instant when it is 2 m above the ground, (2)
- (b) the total time between the instant when the stone is projected from O and the instant when it first strikes the ground. (4)

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January 2021 Q1 — detailed solution

Ground impact

ALL
6 marks**Find displacement for part a**

The stone starts 5 m above the ground. When it is 2 m above the ground, its displacement from O is -3 m.

Use SUVAT for speed

Taking upward as positive, $u = 20$, $a = -9.8$ and $s = -3$. Use $v^2 = u^2 + 2as$: $v^2 = 20^2 + 2(-9.8)(-3) = 458.8$.

State the speed

$v = 21.4 \text{ m s}^{-1}$ to 3 significant figures.

Set up the ground equation

At the ground, displacement from O is -5 m. Use $s = ut + \frac{1}{2}at^2$: $-5 = 20t - 4.9t^2$.

Solve for time

$4.9t^2 - 20t - 5 = 0$. The positive root is $t = 4.3179\dots$, so the total time is 4.32 s.

Final answer (a) 21.4 m s^{-1} . (b) 4.32 s approximately.

January 2022 Q4 — complete original question

4. At time $t = 0$, a small ball is projected vertically upwards from a point A which is 24.5 m above the ground. The ball first comes to instantaneous rest at the point B , where $AB = 19.6$ m and first hits the ground at time $t = T$ seconds.

The ball is modelled as a particle moving freely under gravity.

- (a) Find the value of T .

(6)

- (b) Sketch a speed-time graph for the motion of the ball from $t = 0$ to $t = T$ seconds.

(No further calculations are needed in order to draw this sketch.)

(2)

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January 2022 Q4 — detailed solution

Ground impact

ALL
8 marks**Find the initial speed**

At the highest point B, the velocity is 0 and $AB = 19.6$ m. Use $v^2 = u^2 + 2as$ upward: $0 = u^2 - 2g(19.6)$.

Calculate u

$u^2 = 2(9.8)(19.6) = 384.16$, so $u = 19.6 \text{ m s}^{-1}$.

Use displacement to the ground

From A to the ground, taking upward as positive, displacement is -24.5 m. Use $s = ut + \frac{1}{2}at^2$:
 $-24.5 = 19.6T - 4.9T^2$.

Solve for T

$4.9T^2 - 19.6T - 24.5 = 0$. Divide by 4.9: $T^2 - 4T - 5 = 0$, so $(T - 5)(T + 1) = 0$. The positive time is $T = 5$ s.

Describe the speed-time graph

The speed starts at 19.6, decreases linearly to 0 at $t = 2$, then increases linearly until the ball hits the ground at $t = 5$. The final speed is 29.4, so the second line is longer than the first.

Final answer (a) $T = 5$ s. (b) Speed-time graph is V-shaped, from speed 19.6 at $t = 0$ to 0 at $t = 2$, then to speed 29.4 at $t = 5$.

Summer 2025 Q3 — complete original question

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

- (a) Find the value of U . (3)
- (b) Find the speed of the particle as it hits the ground at B . (3)
- (c) Sketch a speed-time graph for the motion of the particle from the instant it leaves A to the instant it reaches B . (No further calculations are required.) (2)

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Summer 2025 Q3 — detailed solution

Ground impact

ALL
8 marks**Part (a): Set up the signed data**Take upward as positive. Then $u = U$, $s = -12 \text{ m}$, $t = 4 \text{ s}$ and $a = -9.8 \text{ m s}^{-2}$.**Part (a): Find the projection speed**Using $s = ut + \frac{1}{2}at^2$, $-12 = 4U + \frac{1}{2}(-9.8)(4^2) = 4U - 78.4$. Hence $4U = 66.4$ and $U = 16.6 \text{ m s}^{-1}$.**Part (b): Find the impact velocity**Using $v^2 = u^2 + 2as$, $v^2 = 16.6^2 + 2(-9.8)(-12) = 510.76$. The particle is moving downward at B, so $v = -22.6 \text{ m s}^{-1}$.**Part (b): State the requested speed**Therefore the impact speed is $|v| = 22.6 \text{ m s}^{-1}$ (about 23 m s^{-1} to 2 significant figures).**Part (c): Locate the vertex**The speed decreases linearly from 16.6 to 0. From $v = u + at$, $0 = 16.6 - 9.8t$, so the minimum point is $(1.69, 0)$. The graph then rises linearly to $(4, 22.6)$.**Part (c): Sketch the speed–time graph**Draw straight segments through $(0, 16.6)$, $(1.69, 0)$ and $(4, 22.6)$. This is a V-shaped speed–time graph; the two gradients have equal magnitude 9.8.**Final answer** (a) $U = 16.6 \text{ m s}^{-1}$ (about 17 m s^{-1} to 2 s.f.). (b) speed = 22.6 m s^{-1} . (c) V-shaped speed–time graph through $(0, 16.6)$, $(1.69, 0)$ and $(4, 22.6)$.

October 2022 Q5 — complete original question

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (a) Find the distance travelled by the ball while its speed is less than 14.7 m s^{-1} (3)
- (b) Find the time for which the ball is moving with a speed of more than 29.4 m s^{-1} (3)
- (c) Sketch a speed-time graph for the motion of the ball from the instant when it is projected from A to the instant when it hits the ground. Show clearly where your graph meets the axes. (3)

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October 2022 Q5 — detailed solution

Time above a level

ALL
9 marks**Find one side of the slow-speed interval**

The speed is less than 14.7 m s^{-1} near the highest point. From speed 14.7 to speed 0 , use $v^2 = u^2 + 2as$: $0 = 14.7^2 - 2gs$.

Find the total distance

This gives $s = 11.025 \text{ m}$ on the way up. The same distance occurs on the way down before the speed reaches 14.7 again, so the total distance is $2s = 22.05 \text{ m}$, about 22.1 m .

Interpret speed greater than 29.4

The ball has speed 29.4 again when it passes A on the way down. It is faster than 29.4 only after passing A downward until it hits the ground.

Find that time

From A to the ground, taking downward as positive, $19.6 = 29.4t + \frac{1}{2}gt^2$.

Solve the time equation

$19.6 = 29.4t + 4.9t^2$, giving the positive root $t = 0.606 \dots \text{ s}$.

Describe the speed-time graph

The graph starts at speed 29.4 when $t = 0$, decreases linearly to speed 0 at $t = 3$, then increases linearly until the ball hits the ground. The final speed is greater than 29.4 , so the right-hand arm ends above the starting height on the speed axis.

Final answer (a) 22.1 m approximately. (b) 0.606 s approximately. (c) V-shaped speed-time graph through $(0, 29.4)$ and $(3, 0)$, ending above 29.4 .

January 2024 Q6 — complete original question

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of V .

(3)

- (b) Find the time taken for the particle to reach B .

(3)

The point C is 10 m vertically above A .

- (c) Find the length of time for which the particle is above C .

(4)

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches B . (No further calculations are required.)

(2)

January 2024 Q6 — detailed solution**ALL**
12 marks**Find the impact speed**

Take upward as positive from A. At B, the displacement is -2.5 m, with $u = 24$ and $a = -9.8$. Use $v^2 = u^2 + 2as$: $V^2 = 24^2 + 2(-9.8)(-2.5) = 625$.

State V

$$V = 25 \text{ m s}^{-1}.$$

Find the time to reach B

The velocity at B is -25 m s^{-1} . Using $v = u + at$, $-25 = 24 - 9.8t$, so $t = 5$ s.

Set up the height C equation

C is 10 m above A. The particle is above C between the two times when $s = 10$: $10 = 24t - 4.9t^2$.

Solve for the two crossing times

$$4.9t^2 - 24t + 10 = 0, \text{ giving } t = 0.459 \dots \text{ and } t = 4.438 \dots$$

Find the time above C

The length of time above C is $4.438 \dots - 0.459 \dots = 3.98 \dots$ s, about 4.0 s.

Describe the speed-time graph

The graph starts at speed 24, decreases linearly to 0 at the highest point, then increases linearly to speed 25 at $t = 5$. It is a V-shaped speed-time graph.

Final answer (a) $V = 25 \text{ m s}^{-1}$. (b) $t = 5$ s. (c) 4.0 s approximately. (d) Speed-time graph is V-shaped, from speed 24 to 0, then to speed 25 at $t=5$.

TOPIC 04 | FAMILY 03

Piecewise one-dimensional motion

WHAT THIS FAMILY ASKS

Join constant-acceleration stages when force, acceleration, parachute state or model changes.

VARIANT ROUTE

V01 Acceleration or force switches value

V02 Compare alternative piecewise motion models

3 classified questions

33 total marks

January 2023 Q5 — complete original question

5.

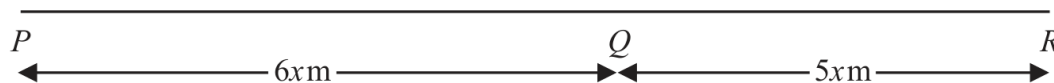


Figure 2

Three points P , Q and R are on a horizontal road where PQR is a straight line.

The point Q is between P and R , with $PQ = 6x$ metres and $QR = 5x$ metres, as shown in Figure 2.

A vehicle moves along the road from P to Q with constant acceleration.

The vehicle is modelled as a particle.

At time $t = 0$, the vehicle passes P with speed $u \text{ ms}^{-1}$

At time $t = 12 \text{ s}$, the vehicle passes Q with speed $2u \text{ ms}^{-1}$

Using the model,

(a) show that $x = 3u$

(2)

As the vehicle passes Q , the acceleration of the vehicle changes instantaneously to 1.5 ms^{-2}

The vehicle continues to move with a constant acceleration of 1.5 ms^{-2} and passes R with speed $3u \text{ ms}^{-1}$

Using the model,

(b) find the value of u ,

(3)

(c) find the distance travelled by the vehicle during the first 14 seconds after passing P

(4)

January 2023 Q5 — detailed solution

Acceleration switch

ALL
9 marks**Use average speed from P to Q**

From P to Q, the speed changes uniformly from u to $2u$ over 12 s. The average speed is $\frac{u + 2u}{2} = \frac{3u}{2}$.

Show $x = 3u$

Distance $PQ = 6x$, so $6x = \frac{3u}{2} \cdot 12 = 18u$. Hence $x = 3u$.

Use Q to R

The distance $QR = 5x = 15u$. The speed changes from $2u$ to $3u$ with acceleration 1.5 m s^{-2} .

Find u

Use $v^2 = u^2 + 2as$: $(3u)^2 = (2u)^2 + 2(1.5)(15u)$. Thus $9u^2 = 4u^2 + 45u$, so $5u^2 = 45u$ and $u = 9$.

Find the first 12 seconds distance

In the first 12 seconds, the distance is $PQ = 6x = 18u = 18(9) = 162 \text{ m}$.

Find the next 2 seconds distance

From $t = 12$ to $t = 14$, the vehicle starts this part at speed $2u = 18$ and accelerates at 1.5 . Distance $= 18(2) + \frac{1}{2}(1.5)(2^2) = 36 + 3 = 39 \text{ m}$.

Add the distances

Total distance in the first 14 seconds is $162 + 39 = 201 \text{ m}$.

Final answer (a) $x = 3u$. (b) $u = 9$. (c) 201 m .

Summer 2024 Q5 — complete original question

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.
- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
 - the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
 - from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
 - the parachute and box continue to fall vertically downwards after the parachute opens
 - the parachute and box are modelled throughout the motion as a particle P of mass 250 kg
- (a) Find the distance fallen by P in the first 5 seconds. (2)
- (b) Find the speed with which P lands on the ground. (7)
- (c) Find the total time from the instant when P is dropped from the helicopter to the instant when P lands on the ground. (3)
- (d) Sketch a speed-time graph for the motion of P from the instant when P is dropped from the helicopter to the instant when P lands on the ground. (2)

Summer 2024 Q5 — detailed solution

Acceleration switch

ALL
14 marks**Find the first distance**

For the first 5 seconds, P falls freely from rest. Using $s = ut + \frac{1}{2}at^2$, $s = 0 + \frac{1}{2}(9.8)(5^2) = 122.5$ m.

Find the speed when the parachute opens

After 5 seconds of free fall, $v = 0 + 9.8(5) = 49 \text{ m s}^{-1}$ downward.

Find the new acceleration

After the parachute opens, weight is $250g = 2450 \text{ N}$ downward and resistance is 3200 N upward.

Taking downward as positive, the resultant is $2450 - 3200 = -750 \text{ N}$, so $a = \frac{-750}{250} = -3 \text{ m s}^{-2}$.

Use the remaining distance

The remaining distance is $520 - 122.5 = 397.5$ m. Using $v^2 = u^2 + 2as$: $v^2 = 49^2 + 2(-3)(397.5) = 16$.

Find the landing speed

$v = 4 \text{ m s}^{-1}$, so P lands with speed 4 m s^{-1} .

Find the time after the parachute opens

Using $v = u + at$, $4 = 49 - 3t$, so $t = 15$ s after the parachute opens.

Find the total time and describe the graph

The total time is $5 + 15 = 20$ s. The speed-time graph rises linearly from 0 to 49 over the first 5 s, then falls linearly to 4 at $t = 20$; it does not meet the time-axis.

Final answer (a) 122.5 m or 123 m. (b) 4 m s^{-1} . (c) 20 s. (d) Speed-time graph: straight line from (0, 0) to (5, 49) then straight line down to (20, 4).

October 2021 Q3 — complete original question

3. A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

Let t seconds be the time that elapses after the car passes A .

In a **first** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until the car reaches the road sign.

- (a) Use this first model to find the speed of the car as it reaches the sign.

(2)

The road sign indicates that the speed limit immediately after the sign is 13 m s^{-1} .

In a **second** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (b) Use this second model to find the value of t at which the car reaches the sign.

(4)

In a **third** model, the car is assumed to move with constant speed 25 m s^{-1} from A until time $t = 0.2$, the car then decelerates uniformly at 6 m s^{-2} until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (c) Use this third model to find the value of t at which the car reaches the sign.

(4)

October 2021 Q3 — detailed solution

Compare alternative models

ALL
10 marks**Use the first model**From A to the sign, $u = 25$, $a = -6$ and $s = 48$. Use $v^2 = u^2 + 2as$.**Find the speed at the sign** $v^2 = 25^2 + 2(-6)(48) = 625 - 576 = 49$, so $v = 7 \text{ m s}^{-1}$.**Second model: time to reach 13**Use $v = u + at$: $13 = 25 - 6t$, so $t = 2 \text{ s}$ to slow to 13 m s^{-1} .**Second model: distance while slowing**Distance while slowing is $\frac{25 + 13}{2} \cdot 2 = 38 \text{ m}$.**Second model: finish time**The remaining distance is $48 - 38 = 10 \text{ m}$ at speed 13. Total time $= 2 + \frac{10}{13} = \frac{36}{13} \text{ s}$.**Third model: first 0.2 seconds**The car first travels at 25 m s^{-1} for 0.2 s , covering 5 m .**Third model: slowing distance**It then takes another 2 s to slow from 25 to 13 , covering 38 m as before. Total covered before constant-speed part is $5 + 38 = 43 \text{ m}$.**Third model: finish time**The remaining distance is $48 - 43 = 5 \text{ m}$ at speed 13. Total time $= 0.2 + 2 + \frac{5}{13} = \frac{168}{65} \text{ s}$.**Final answer** (a) 7 m s^{-1} . (b) $t = \frac{36}{13} \text{ s} \approx 2.77 \text{ s}$. (c) $t = \frac{168}{65} \text{ s} \approx 2.58 \text{ s}$.

TOPIC 04 | FAMILY 04

Rebound and repeated-impact kinematics

WHAT THIS FAMILY ASKS

Track vertical flights separated by instantaneous impacts that change the velocity.

VARIANT ROUTE

V01 Repeated rebounds or speed ratio

V02 Rebound linked to impulse

V03 Time above a height across rebounds

3 classified questions

29 total marks

October 2019 Q2 — complete original question

2. A small ball is released from rest from a point that is 40 m above horizontal ground. The ball bounces on the ground and rebounds vertically. Each time the ball bounces on the ground, the speed of the ball is instantaneously reduced by 50%. The ball is modelled as a particle moving freely under gravity, from the instant when it is released until it first hits the ground, and between each successive bounce.
- (a) Find the time from the instant when the ball is released from rest to the instant when it hits the ground for the second time. (5)
- (b) Find the maximum height reached by the ball above the ground after the ball's third bounce. (4)

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October 2019 Q2 — detailed solution

Repeated rebounds

ALL
9 marks**Find the time to the first hit**

The ball is released from rest and falls 40 m. Use $s = ut + \frac{1}{2}gt^2$: $40 = \frac{1}{2}gt_1^2$. Thus $t_1 = \sqrt{\frac{80}{g}} = \frac{20}{7}$ s.

Find the speed just before the first bounce

Using $v^2 = u^2 + 2gs$, $v_1^2 = 2g(40)$, so $v_1 = 28 \text{ m s}^{-1}$.

Use the rebound speed

After the first bounce the speed is reduced by 50%, so the upward speed is 14 m s^{-1} .

Find the time until the second hit

The time to go up and return to the ground is $\frac{2(14)}{g} = \frac{20}{7}$ s.

Find the total time

Total time to the second hit is $\frac{20}{7} + \frac{20}{7} = \frac{40}{7} = 5.71$ s approximately.

Find the speed after the third bounce

After each bounce, the speed is halved. The upward speeds after the first, second and third bounces are 14, 7 and 3.5 m s^{-1} .

Find the height after the third bounce

At the top after the third bounce, $0 = (3.5)^2 - 2gh$. Therefore $h = \frac{(3.5)^2}{2g} = 0.625$ m.

Final answer (a) 5.71 s approximately. (b) 0.625 m.

October 2024 Q7 — complete original question

- 7 At time $t = 0$, a small ball A is projected vertically upwards with speed 8 m s^{-1} from a fixed point on horizontal ground.

The ball hits the ground again for the first time at time $t = T_1$ seconds.

Ball A is modelled as a particle moving freely under gravity.

- (a) Show that $T_1 = 1.63$ to 3 significant figures.

(2)

After the first impact with the ground, A rebounds to a height of 2 m above the ground.

Given that the mass of A is 0.1 kg,

- (b) find the magnitude of the impulse received by A as a result of its first impact with the ground.

(5)

At time $t = 1$ second, another small ball B is projected vertically upwards from another point on the ground with speed 5 m s^{-1}

Ball B is modelled as a particle moving freely under gravity.

At time $t = T_2$ seconds ($T_2 > 1$), A and B are at the same height above the ground for the first time.

- (c) Find the value of T_2

(4)

October 2024 Q7 — detailed solution

Rebound and impulse

ALL
11 marks**Find the first flight time**

For ball A, take upward as positive. Returning to the ground gives $0 = 8T_1 - 4.9T_1^2$. The non-zero solution is $T_1 = \frac{8}{4.9} = 1.63$ s to 3 significant figures.

Find the rebound speed

After impact, A rebounds to a maximum height of 2 m. At the top, $v = 0$, so $0 = u^2 - 2g(2)$. Hence $u = \sqrt{4g} = 6.261$ m s⁻¹.

Use impulse as change in momentum

Just before the first impact, A has velocity -8 m s⁻¹. Just after impact, it has velocity $+6.261$ m s⁻¹. With mass 0.1 kg, $I = 0.1[6.261 - (-8)]$.

Calculate the impulse

$I = 1.426 \dots$ N s, so the impulse is 1.43 N s to 3 significant figures.

Write the height of A before first impact

For the first meeting with B, A is still in its first flight. Its height at time T_2 is $h_A = 8T_2 - 4.9T_2^2$.

Write the height of B

B starts at time $t = 1$, so at time T_2 it has been moving for $T_2 - 1$ seconds. Its height is $h_B = 5(T_2 - 1) - 4.9(T_2 - 1)^2$.

Set the heights equal

$8T_2 - 4.9T_2^2 = 5(T_2 - 1) - 4.9(T_2 - 1)^2$. Expanding and simplifying gives $6.8T_2 = 9.9$.

Find T_2

$T_2 = \frac{9.9}{6.8} = 1.456 \dots$, so $T_2 = 1.46$ s to 3 significant figures.

Final answer (a) $T_1 = 1.63$ s. (b) $I = 1.43$ N s. (c) $T_2 = 1.46$ s.

Summer 2026 Q4 — complete original question

4. The point O is 28.4 m above a horizontal plane. A small ball is projected vertically upwards from O with speed $U \text{ m s}^{-1}$ and first hits the plane with speed 24 m s^{-1}

Air resistance is assumed to be negligible.

- (a) Find the value of U .

(3)

Each time the ball hits the plane, the speed of the ball is instantaneously halved.

- (b) Find the **total** time, **after the ball has hit the plane for the first time**, for which the ball is at least 4 m above the plane.

(6)

DR ESLAM AHMED

Summer 2026 Q4 — detailed solution

Rebound height threshold

ALL
9 marks**Part (a): Choose a sign convention**

Take upward as positive, so $s = -28.4$, $a = -9.8$ and the impact velocity is -24 . Use $v^2 = u^2 + 2as$ with initial speed U .

Part (a): Solve for the projection speed

$24^2 = U^2 + 2(9.8)(28.4)$, giving $U^2 = 19.36$. Since U is a speed, $U = 4.4 \text{ m s}^{-1}$.

Part (b): Model the first rebound

After the first impact the upward speed is 12 m s^{-1} . If t is measured from that impact, the height above the plane is $y = 12t - 4.9t^2$.

Part (b): Find the threshold times

At height 4 metres, $4.9t^2 - 12t + 4 = 0$. The roots are $0.398022 \dots$ and $2.050957 \dots$ seconds.

Part (b): Calculate the first-rebound duration

The ball is at least 4 metres high between the roots, for $2.050957 \dots - 0.398022 \dots = 1.652935 \dots$ seconds.

Part (b): Exclude all later rebounds

The second rebound starts at 6 m s^{-1} and reaches only $6^2 / (2 \times 9.8) = 1.8367 \dots$ metres. Every later rebound is lower, so none contributes more time above 4 metres.

Final answer (a) $U = 4.4 \text{ m s}^{-1}$. (b) 1.65 s .

TOPIC 04 | FAMILY 05

Meeting of constant-acceleration particles

WHAT THIS FAMILY ASKS

Find when two scalar position functions are equal and extract event velocities.

VARIANT ROUTE

V01 Vertical projected-particle meeting

2 classified questions

20 total marks

January 2020 Q3 — complete original question

3. A particle, P , is projected vertically upwards with speed U from a fixed point O . At the instant when P reaches its greatest height H above O , a second particle, Q , is projected with speed $\frac{1}{2}U$ vertically upwards from O .
- (a) Find H in terms of U and g . (2)
- (b) Find, in terms of U and g , the time between the instant when Q is projected and the instant when the two particles collide. (6)
- (c) Find where the two particles collide. (3)

DR ESLAM AHMED

January 2020 Q3 — detailed solution

Vertical meeting of particles

ALL
11 marks**Find the greatest height**

For P, at greatest height the speed is zero. Use $v^2 = u^2 + 2as$ with $u = U$, $v = 0$, $a = -g$ and $s = H$:
 $0 = U^2 - 2gH$. Hence $H = \frac{U^2}{2g}$.

Start timing when Q is projected

At this instant P is at height H and has zero velocity. After t seconds, P has fallen $\frac{1}{2}gt^2$, so its height above O is $H - \frac{1}{2}gt^2$.

Write Q's height

Q is projected upwards from O with speed $\frac{1}{2}U$, so its height after t seconds is $\frac{1}{2}Ut - \frac{1}{2}gt^2$.

Equate the heights

At collision, $H - \frac{1}{2}gt^2 = \frac{1}{2}Ut - \frac{1}{2}gt^2$. The gravity terms cancel, so $H = \frac{1}{2}Ut$.

Find the time

Substitute $H = \frac{U^2}{2g}$: $\frac{U^2}{2g} = \frac{1}{2}Ut$. Therefore $t = \frac{U}{g}$.

Find the collision position

Substitute $t = \frac{U}{g}$ into P's height: $H - \frac{1}{2}g\left(\frac{U}{g}\right)^2 = \frac{U^2}{2g} - \frac{U^2}{2g} = 0$. Thus they collide at O.

Final answer (a) $H = \frac{U^2}{2g}$. (b) $t = \frac{U}{g}$. (c) The particles collide at O.

October 2025 Q4 — complete original question

4. The point A is 10 m above horizontal ground.
At time $t = 0$, a particle P is projected vertically upwards with speed 5 m s^{-1} from A .
Particle P moves freely under gravity.

(a) Find the greatest height above A reached by P .

(3)

The point B is on the ground, vertically below A .

At time $t = 1$ second, a particle Q is projected vertically upwards with speed 7 m s^{-1} from B .

Particle Q moves freely under gravity.

Particles P and Q collide at time $t = T$ seconds.

(b) Find the value of T .

(4)

(c) Find the speed of P at the instant immediately before the particles collide.

(2)

DR ESAM AHMED

October 2025 Q4 — detailed solution

Vertical meeting of particles

ALL
9 marks**Part (a): Set the vertical model**

Take upward as positive. At maximum height, $v = 0$; initially $u = 5 \text{ m s}^{-1}$ and $a = -9.8 \text{ m s}^{-2}$. Let s be the displacement above A.

Part (a): Find the maximum height

Using $v^2 = u^2 + 2as$, $0^2 = 5^2 + 2(-9.8)s$, so $19.6s = 25$ and $s = \frac{25}{19.6} = 1.276 \dots = 1.28 \text{ m}$ to 3 significant figures.

Part (b): Write P's height

Measure height from ground level and time from P's projection. Since A is 10 m above the ground, $s_P = 10 + 5t - 4.9t^2$.

Part (b): Write Q's height

Q is projected from ground level at $t = 1$ with speed 7 m s^{-1} , so $s_Q = 7(t - 1) - 4.9(t - 1)^2 = -4.9t^2 + 16.8t - 11.9$.

Part (b): Solve for the meeting time

At collision $s_P = s_Q$. Hence $10 + 5T - 4.9T^2 = -4.9T^2 + 16.8T - 11.9$, so $21.9 = 11.8T$ and $T = 1.855 \dots = 1.86 \text{ s}$ to 3 significant figures.

Part (c): Find P's speed

At $T = 1.86 \text{ s}$, $v_P = 5 - 9.8(1.86) = -13.228 \dots \text{ m s}^{-1}$. The negative sign means P is moving downward, so its speed is $|v_P| = 13.2 \text{ m s}^{-1}$.

Final answer (a) maximum height above A = 1.28 m. (b) $T = 1.86 \text{ s}$. (c) speed of P = 13.2 m s^{-1} .